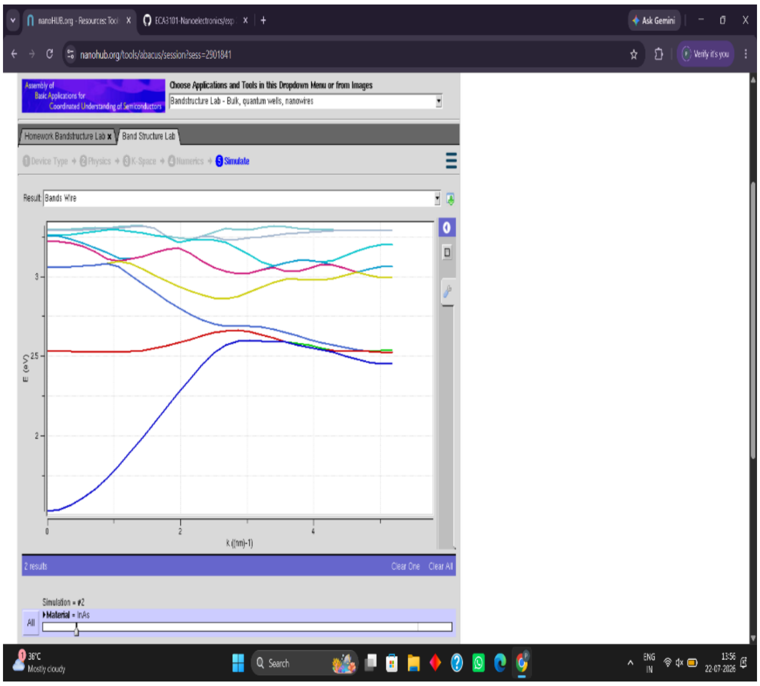
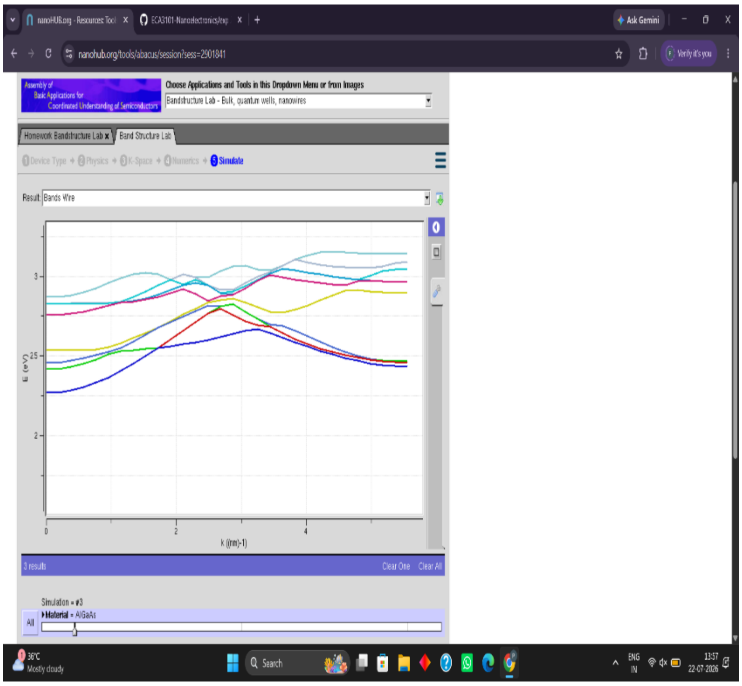
Exp. 5. Band Structure Simulation for Graphene and CNTs using NanoHub

Case Study 1: Simulate the band-structure of InAs circular nanowire



The graph shows the **electronic band structure**, representing energy levels as a function of the wave vector k. The different colored curves indicate the **dispersion of various energy bands** within the simulated nanostructure.

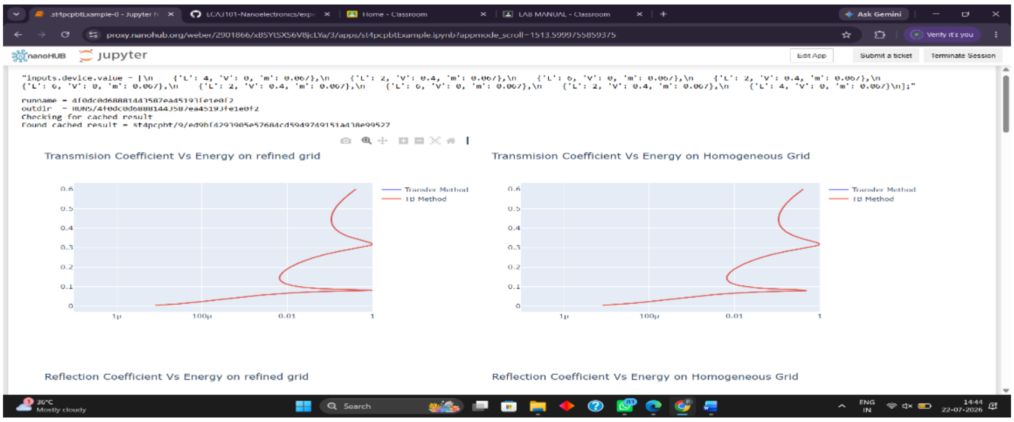
Case Study 2: Simulate the band-structure of AlGaAs circular nanowire



This graph presents another **band structure calculation** with multiple energy bands varying along the k-vector. The curves show how the **energy states change with electron momentum** across the simulated system.

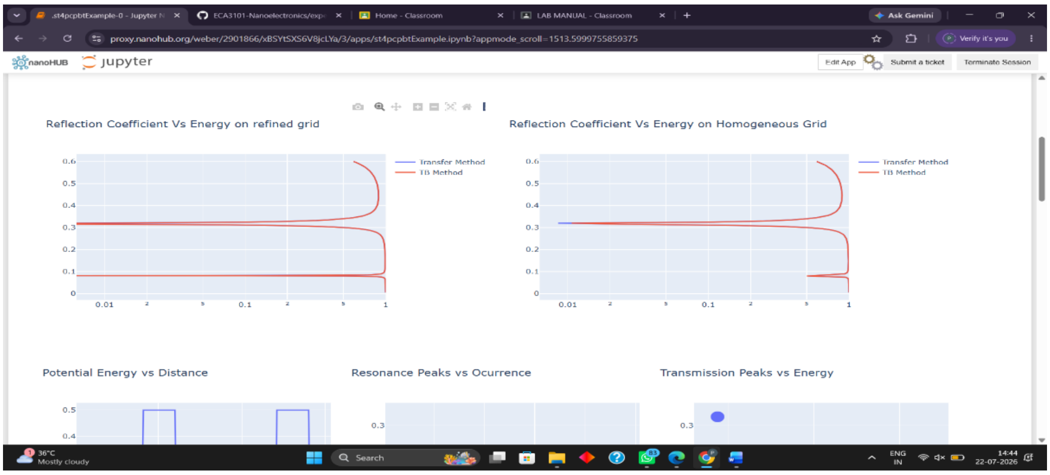
Exp 6 Demonstration of quantum tunneling

Transmission Coefficient vs Energy



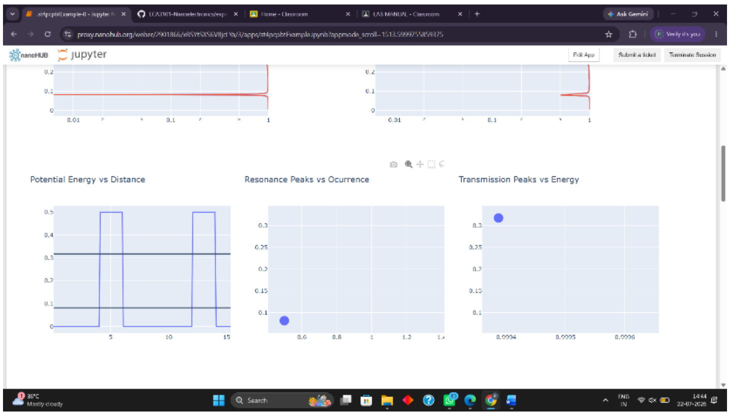
The graph shows the variation of the **transmission coefficient with energy** for both refined and homogeneous grids. The transfer method and TB method show **closely matching results**, indicating good agreement between the two approaches.

Reflection Coefficient and Other Results



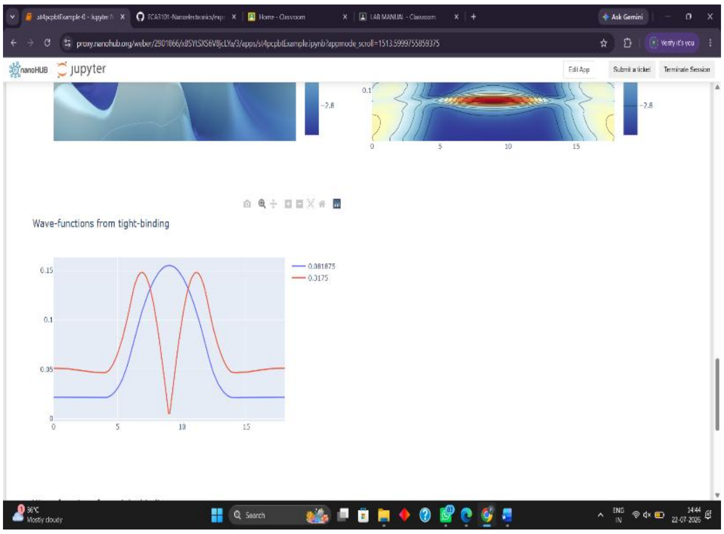
The reflection coefficient plots show the **variation of reflected electron waves with energy** for refined and homogeneous grids. The refined and homogeneous grid results are very similar, confirming the **consistency and accuracy of the simulation**

Potential Energy, Resonance Peaks and Transmission



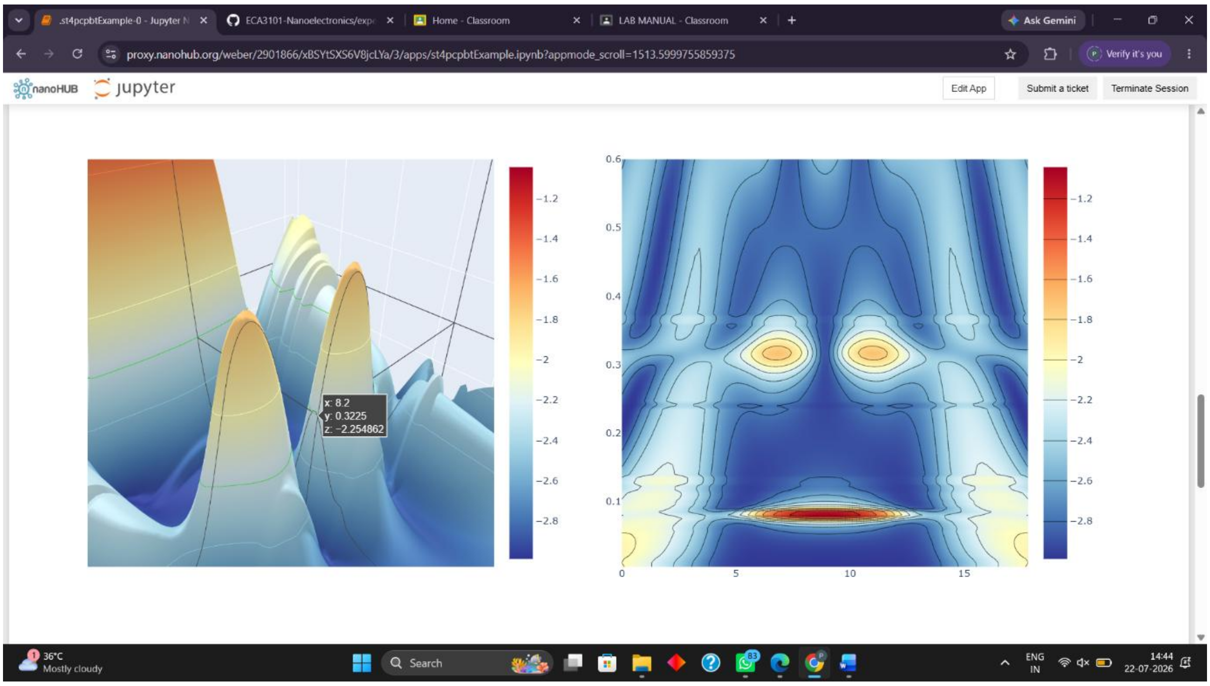
The potential energy plot shows how the **energy barrier changes with distance**, forming distinct regions that affect electron transport. The resonance plot indicates **specific energy levels where resonant transport occurs** within the simulated structure.

3D and Contour Energy Distribution



The 3D surface plot represents the **spatial variation of energy or potential distribution** in the simulated nanostructure. The contour plot provides a **2D view of energy distribution**, clearly showing regions of high and low energy.

Wave Functions from Tight-Binding Method

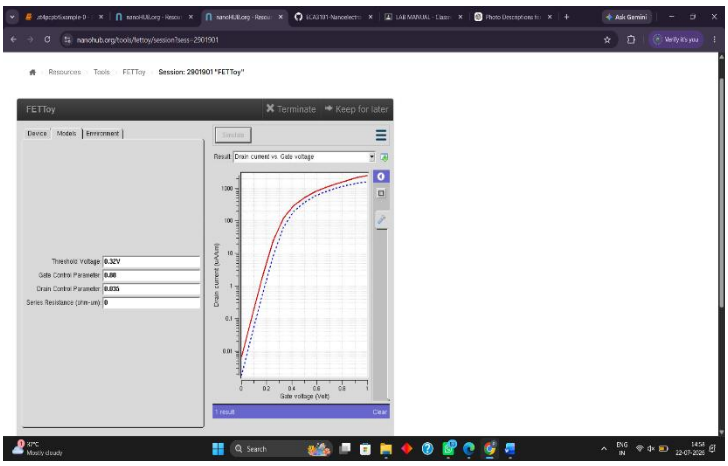


The graph shows the **wave functions obtained using the tight-binding method** for two different energy values. The wave functions exhibit distinct peaks and a central variation, representing the **quantum behavior of electrons** in the nanostructure.

Exp. 7. MOSFET Scaling Analysis and Subthreshold Swing Calculation using

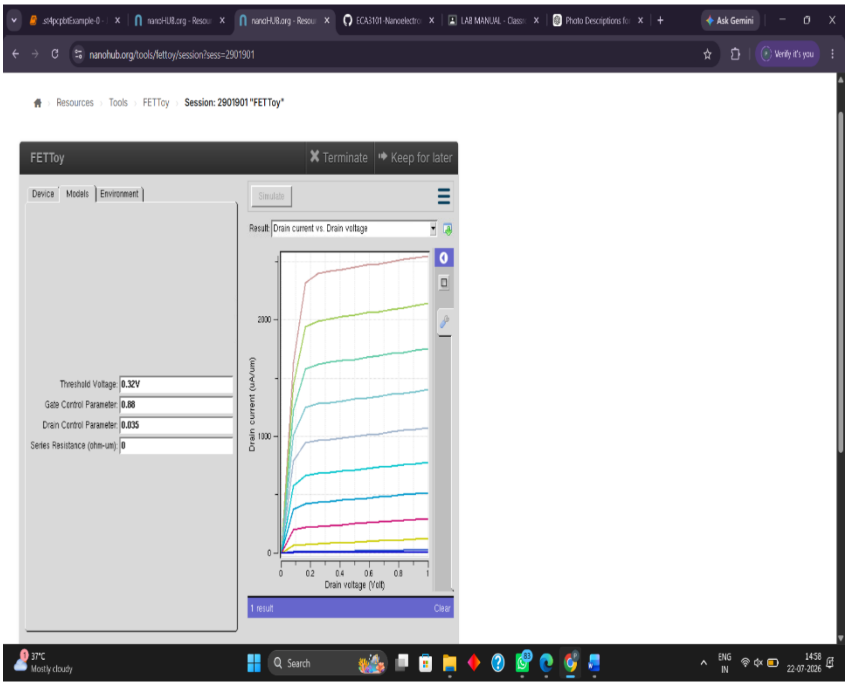
NanoHub

CASE STUDY 1 DRAIN CURRENT VS GATE VOLTAGE



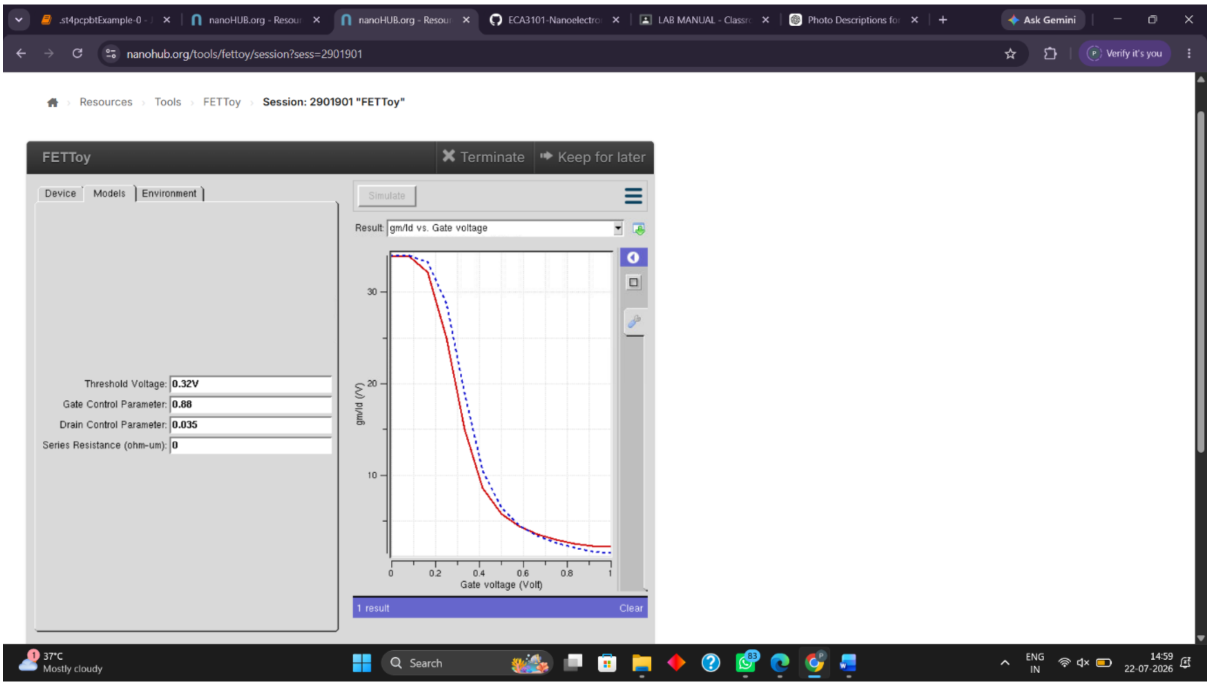
The graph shows the variation of **drain current with gate voltage** for the simulated FET device. As the gate voltage increases, the drain current rises significantly, indicating **enhanced channel conduction**.

CASE STUDY 2 DRAIN CURRENT VS DRAIN VOLTAGE



The graph represents the **drain current as a function of drain voltage** for different gate-control conditions. Drain current initially increases rapidly and then approaches a **nearly constant saturation region**.

CASE STUDY 3 gm/id vs gate voltage



The graph shows the variation of **gm/Id with gate voltage**, indicating the efficiency of gate control over the drain current. The gm/Id value decreases sharply as the gate voltage increases, showing a transition from **strong gate control to higher conduction**.